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Problem

- Standard few-shot prompts map questions directly to answers and often fail on multi-step reasoning.
- Training on large rationale datasets can teach intermediate reasoning but is expensive to annotate.
- Can an off-the-shelf model infer the reasoning pattern from only a few question-rationale-answer examples?

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Emergence

SCALE UNLOCKS THE PROMPT

- Small models often generate fluent but illogical chains and may perform worse than direct answering.
- Reliable gains emerge around 100B parameters; PaLM 540B's reasoning curves rise where standard prompting stays flat.
- Ablations show equations alone are insufficient on semantically hard GSM8K; natural-language decomposition is essential.

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Method

SHOW INTERMEDIATE STEPS

- Augment each in-context exemplar with a coherent natural-language chain leading from input to output.
- At test time the model generates its own chain before stating the final answer; no weights are updated.
- Intermediate text decomposes problems, allocates more tokens/computation to harder cases and exposes debuggable steps.

CHAIN OF THOUGHT

Chain-of-Thought Prompting Elicits Reasoning

Reasoning in Large Language Models | NeurIPS 2022

A few worked examples that include intermediate natural-language reasoning can unlock multi-step arithmetic, commonsense and symbolic capabilities

already latent in sufficiently large language few-shot prompting reasoning models. emergence

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Evidence

EIGHT EXAMPLES, LARGE GAINS

- PaLM 540B with eight CoT exemplars reaches about 57% GSM8K versus 18% with standard prompting.
- It surpasses finetuned GPT-3 with a verifier and produces strong gains across arithmetic and commonsense tasks.
- On symbolic tasks, CoT approaches 100% in-domain and improves generalization to sequences longer than demonstrations.

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Experimental scope

THREE FORMS OF REASONING

- Arithmetic: GSM8K, SVAMP, ASDiv, AQuA and MAWPS using one shared set of eight exemplars.
- Commonsense: five benchmarks spanning physical, social and explanatory knowledge.
- Symbolic: last-letter concatenation and coin-flip tasks test abstract manipulation and longer out-of-distribution chains.

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Limits & meaning

- Readable chains do not prove the network internally reasoned that way, and a plausible chain can still be wrong.
- Manual rationale creation can be costly at scale; benefits originally appeared only in expensive large models.
- The central insight is that prompting changes the observable capability frontier—standard prompts are only a lower bound.